Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.399589 [min 0.399589 , max 0.399589]
Von Neumann entropy : mean = 1.309036 [min 1.309036 , max 1.309036]
Rank of rho : mean = 11.00 [min 11 , max 11]
C_GME (best estimate) : mean = 0.257535 [min 0.228021 , max 0.286142]

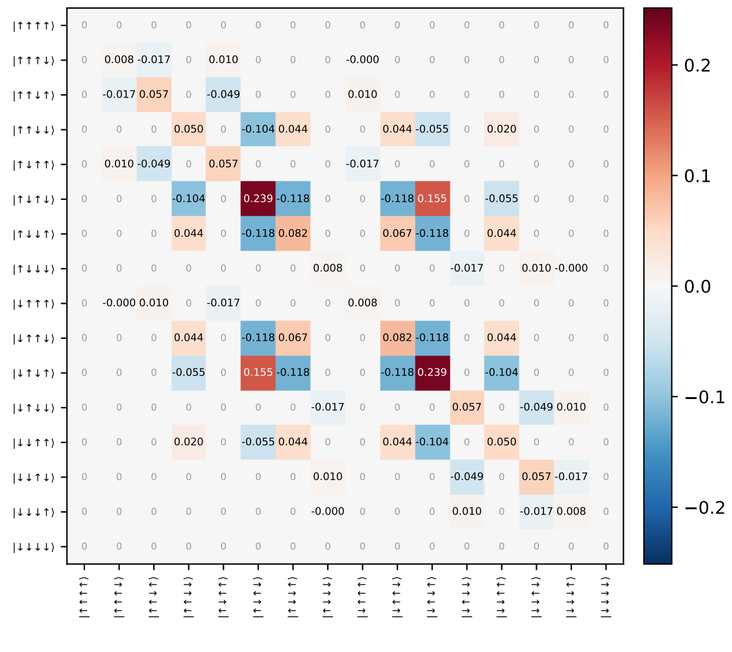
Summary: All 4-Body RDM Clusters

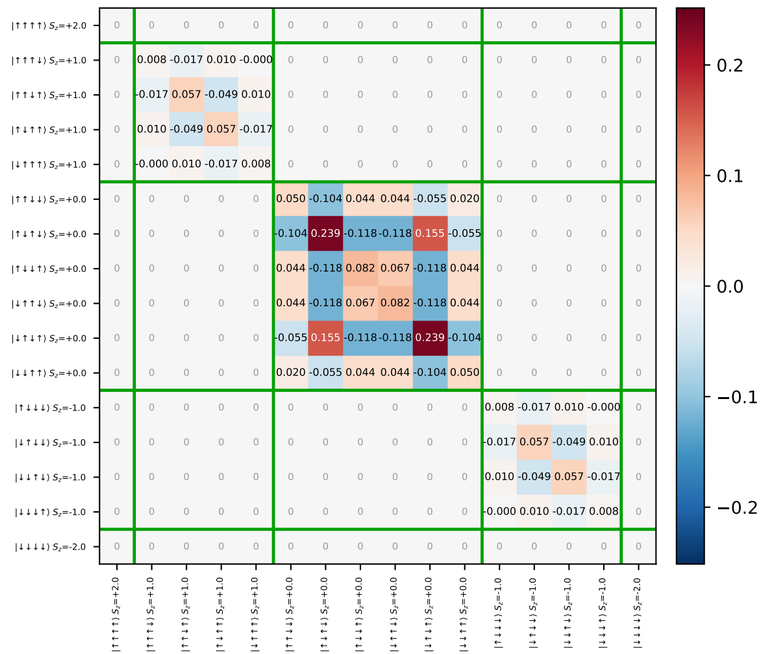
System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-3.668706

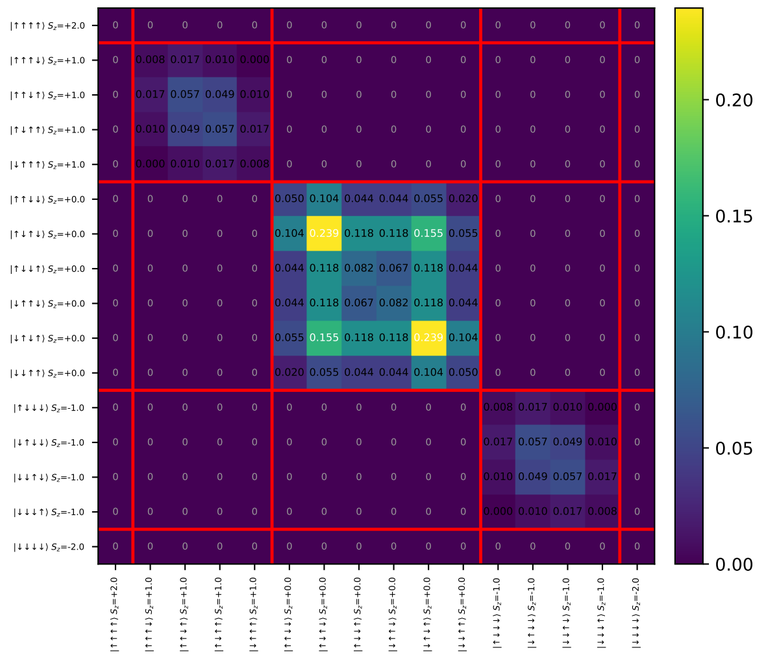
Sites	Trace	Purity	Entropy	C_GME	C_LB	C_gcg	C_stiefel_gcg	C_stiefel	C_UB	Source	Status
(0, 1, 2, 3)	1.000000	0.399589	1.309036	–	0.052886	0.162811	0.161622	0.283579	0.627018	S:PKL G:PKL SG:PKL	MIXED
(1, 2, 3, 4)	1.000000	0.399589	1.309036	–	0.052886	0.163131	0.163019	0.234942	0.627657	S:PKL G:PKL SG:PKL	MIXED
(2, 3, 4, 5)	1.000000	0.399589	1.309036	–	0.052886	0.161358	0.162910	0.237476	0.627656	S:PKL G:PKL SG:PKL	MIXED
(3, 4, 5, 0)	1.000000	0.399589	1.309036	–	0.052886	0.159197	0.162386	0.275049	0.626890	S:PKL G:PKL SG:PKL	MIXED
(4, 5, 0, 1)	1.000000	0.399589	1.309036	–	0.052886	0.161901	0.158967	0.228021	0.626846	S:PKL G:PKL SG:PKL	MIXED
(5, 0, 1, 2)	1.000000	0.399589	1.309036	–	0.052886	0.160902	0.161009	0.286142	0.627327	S:PKL G:PKL SG:PKL	MIXED

Legend:
  Trace = Tr(rho) | Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_gcg = unitary optimization - exponential-map GCG | BEST = selected estimate
  Source: S=Stiefel, G=GCG, SG=Stiefel-GCG; PKL=reused from pkl, NEW=recomputed, MISS->NEW=pkl missing so recomputed
  C_UB = Upper bound (eigendecomp) | Status: GME=has GME | NO GME=separable | MIXED=mixed state

4-Body RDM (Sites 0,1,2,3) — same matrix in all three views







Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -1.451710e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

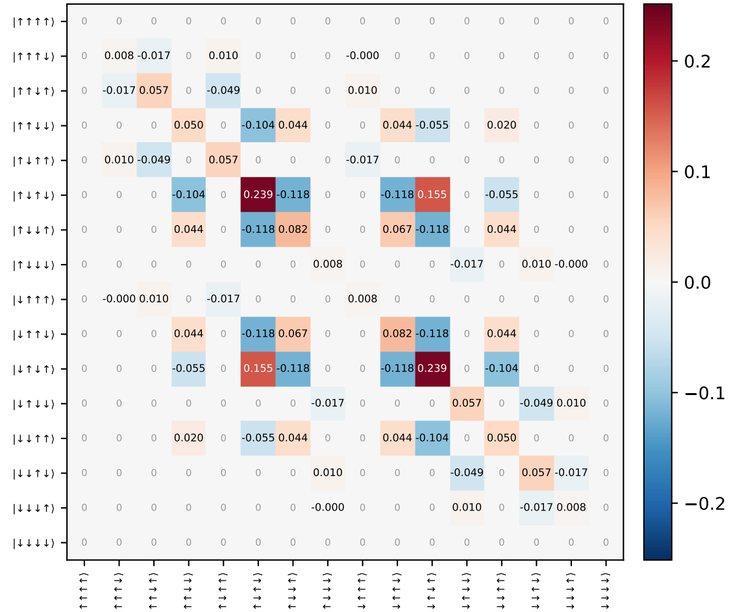
```
Convex Roof Analysis
====================================

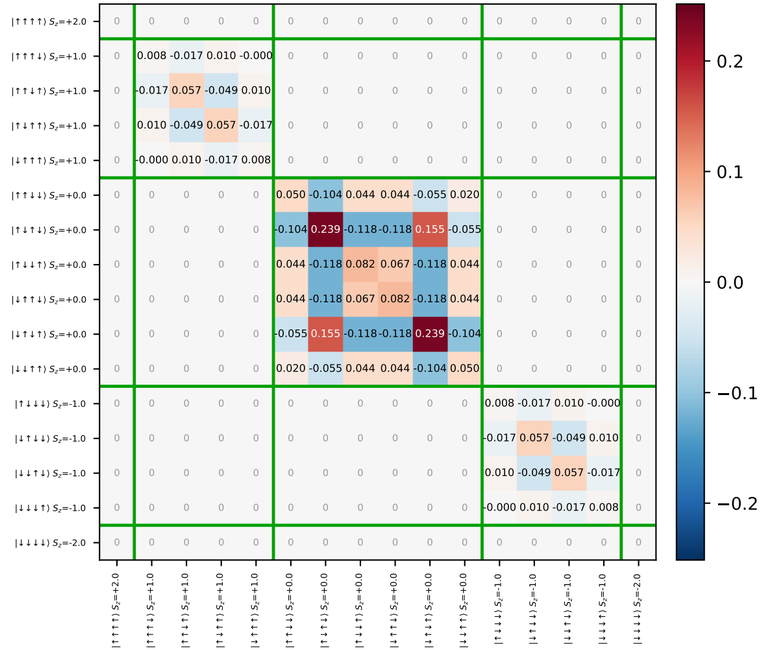
  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.28357874   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62701797

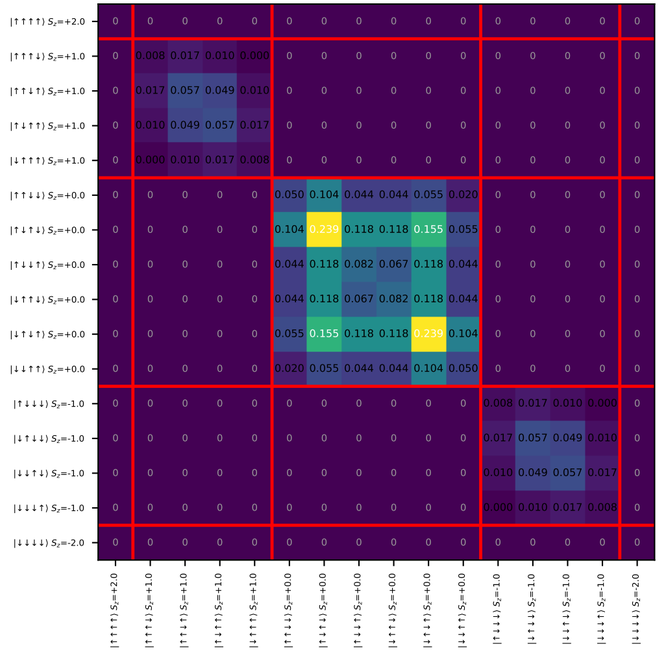
  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```

4-Body RDM (Sites 1,2,3,4) — same matrix in all three views







Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -7.061341e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

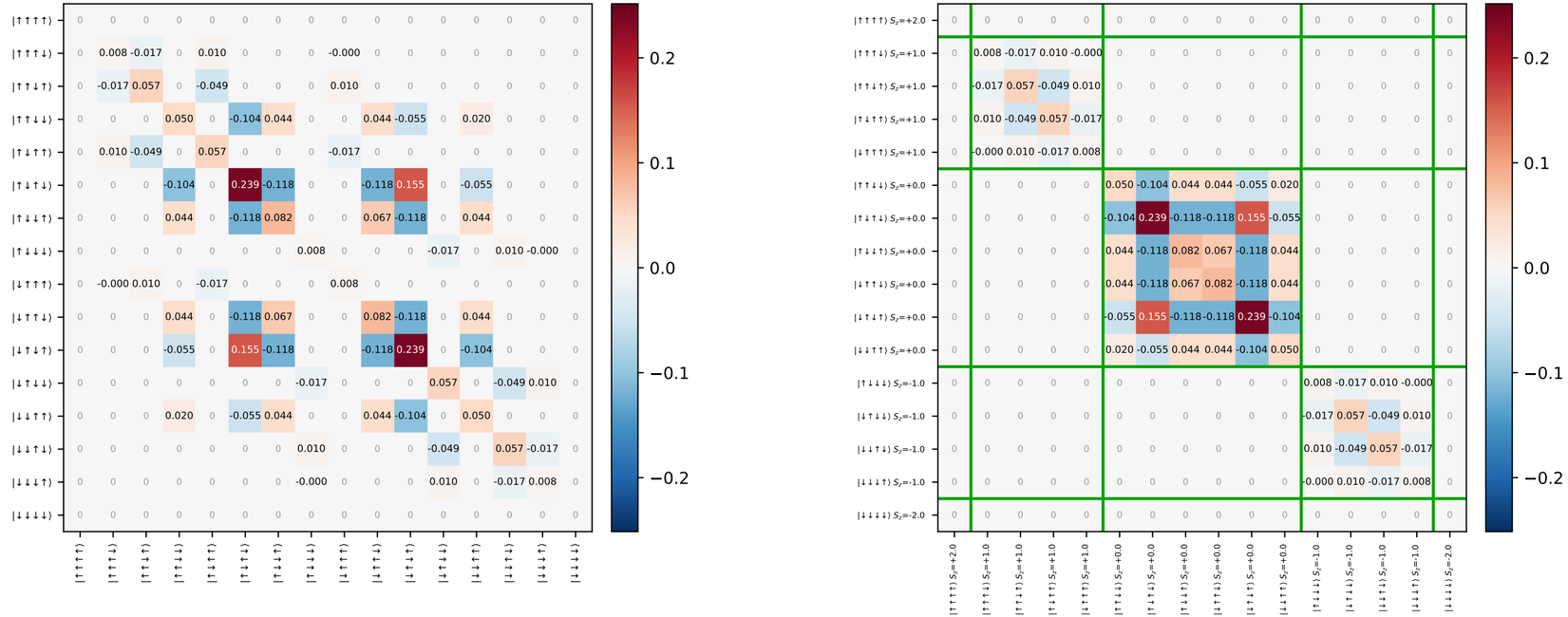
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23494228   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62765726

  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```

4-Body RDM (Sites 2,3,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -1.918754e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
==================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

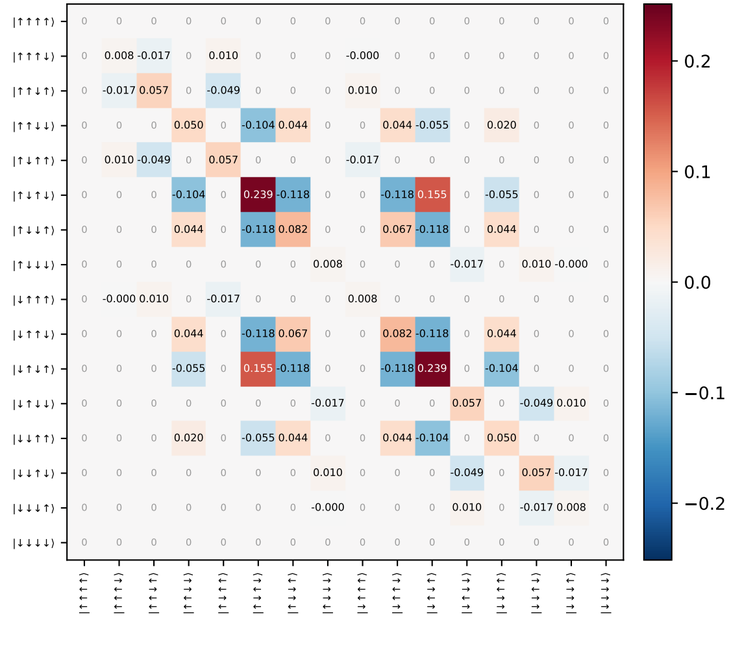
```
Convex Roof Analysis
==================================

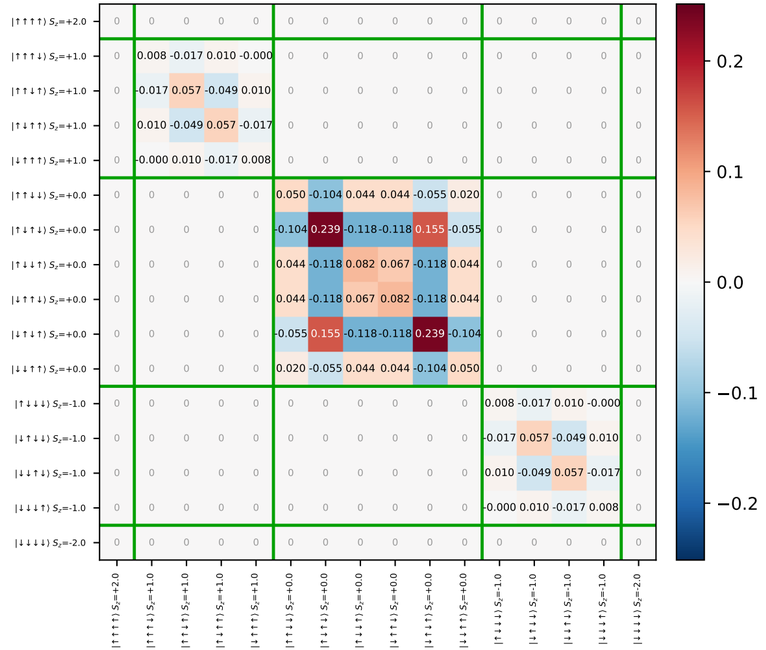
  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23747610   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62765569

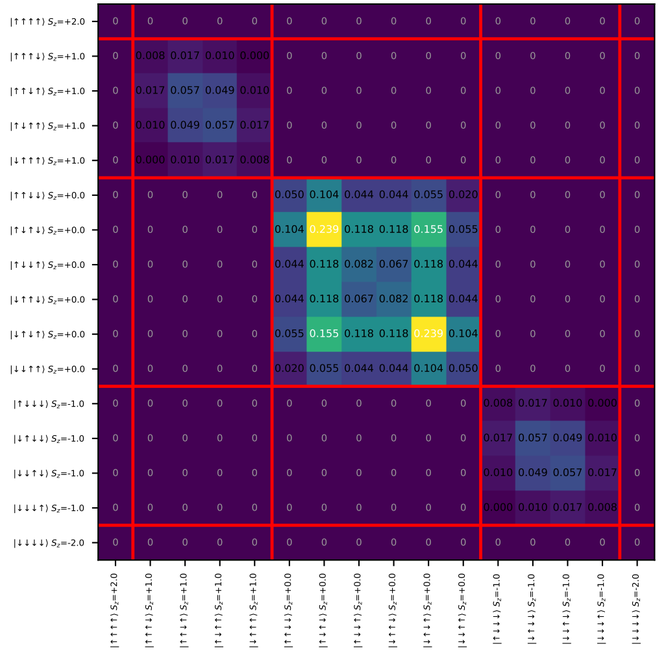
  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```

4-Body RDM (Sites 3,4,5,0) — same matrix in all three views







Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 3,4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -2.868703e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

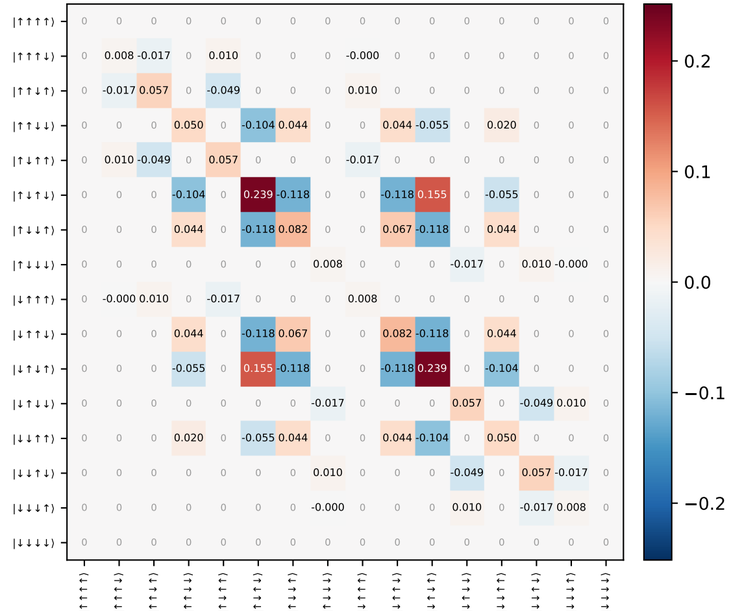
```
Convex Roof Analysis
====================================

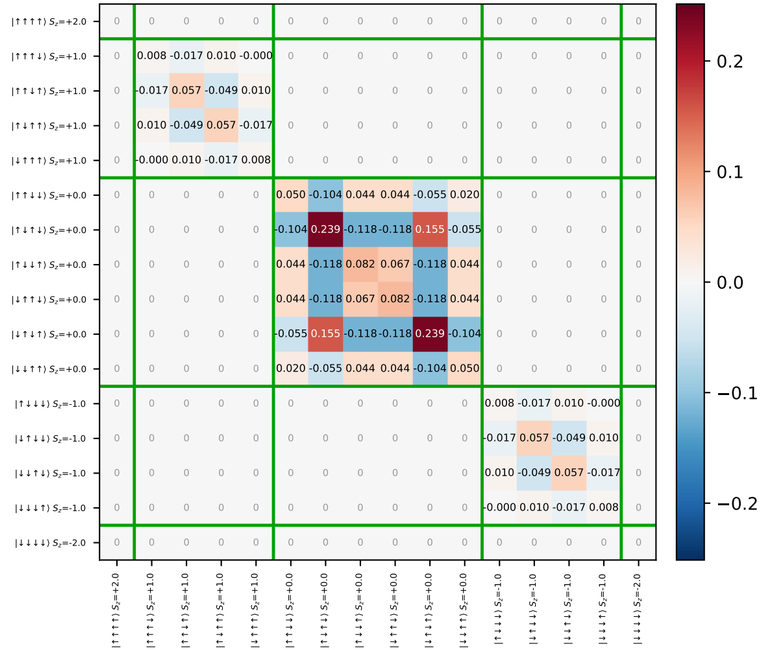
  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.27504866   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62688956

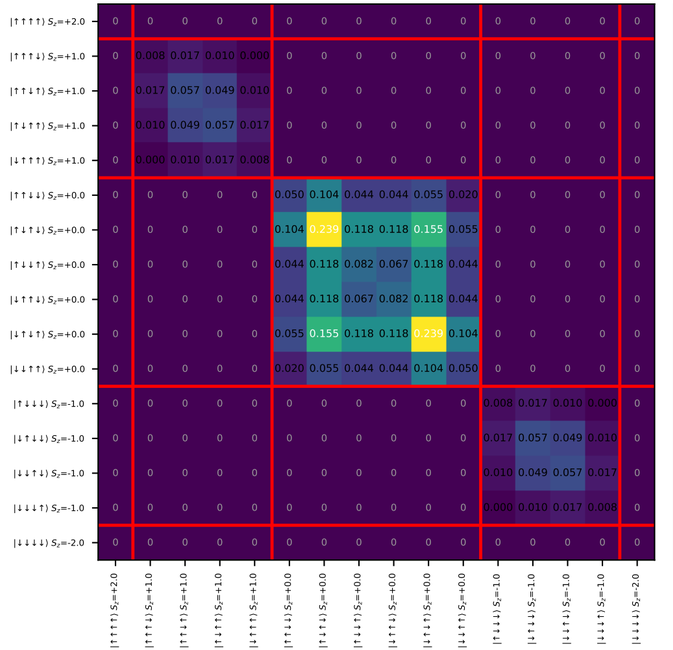
  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```

4-Body RDM (Sites 4,5,0,1) — same matrix in all three views







Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 4,5,0,1)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

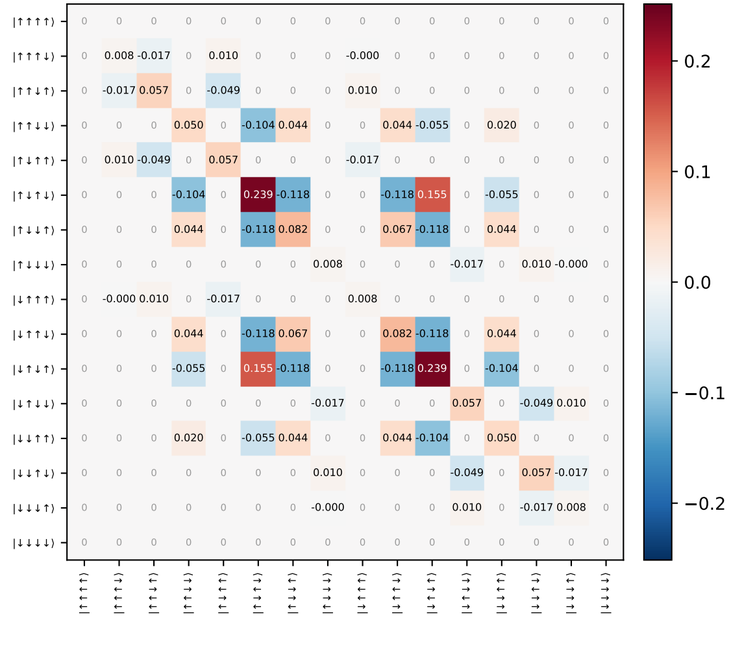
```
Convex Roof Analysis
===================================

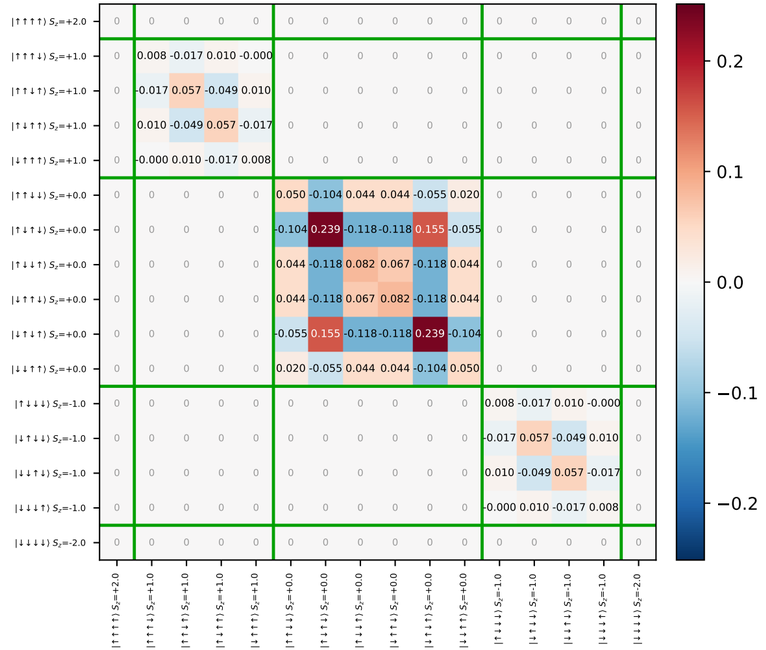
  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.22802098   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62684617

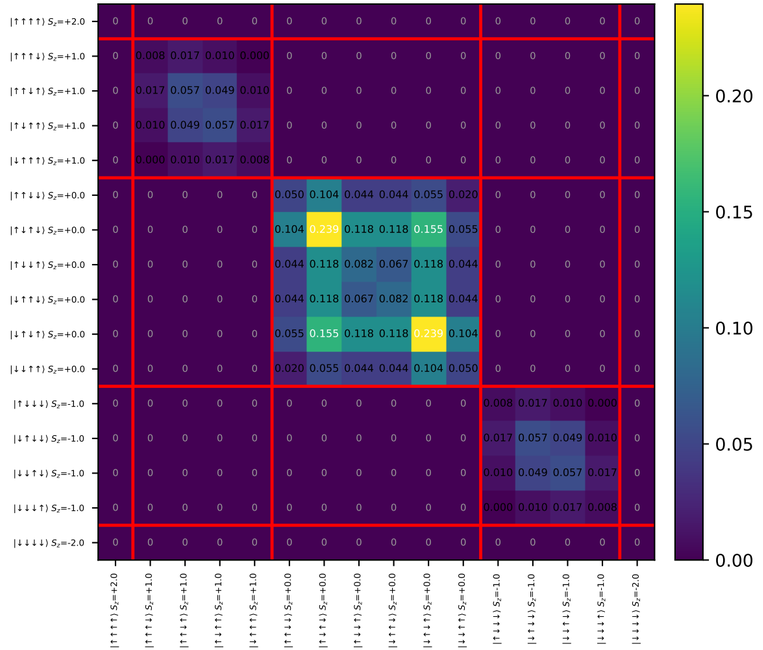
  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```

4-Body RDM (Sites 5,0,1,2) — same matrix in all three views







Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown) | max cross-S_z element = 0.0e+00

Analysis Summary - 4-Body RDM (Sites 5,0,1,2)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -4.728438e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
==================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.28614178   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62732692

  Status: nONE

  Solver source:
    stiefel: PKL  gcg: PKL  stiefel_gcg: PKL
```